

Artem Semidetnov

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Education

BSc	Saint-Petersburg State University , Mathematics	2021 – 2025
BSc	Neapolis University Of Pafos , Applied Computer Science JetBrains Program	2022 – 2024
MSc	Geneva University , Mathematics	2025 – 2027

Research Interests

- Categorical homotopy theory; simplicial methods and operads; homotopy type theory and computer-formalized approaches; geometric and combinatorial group theory (stable commutator length, one-relator groups).

Publications

Operadic double dual (preprint in preparation)

Arshak Aivazian, Emmanuel Farjoun, Vasilii Ionin, **Artem Semidetnov**

The operad associated to a crossed simplicial group [↗](#)

Dec 2025

Artem Semidetnov · [doi:10.48550/arXiv.2512.14498](https://doi.org/10.48550/arXiv.2512.14498) [↗](#)

On the geometry of free nilpotent groups [↗](#)

May 2021

Artem Semidetnov, Ruslan Magdiev · [doi:10.48550/arXiv.2106.00095](https://doi.org/10.48550/arXiv.2106.00095) [↗](#)

AI-Assisted Mathematical Research

- An explicit non-elementary matrix over a bivariate Laurent polynomial ring [↗](#) — Vasilii Ionin, **Artem Semidetnov** (Sept 2026).
- On Some More Problems from the Kourouva Notebook [↗](#) — Vasily Ionin, **Artem Semidetnov** (Aug 2026).
- The stable commutator length of a relator is not a one-relator group invariant [↗](#) — **Artem Semidetnov** (Aug 2026).

Work Experience

JetBrains, Research Intern in HoTT and Dependent Types Lab

Remote

- Developed AI features for the Arend language including MCP servers for typechecking, IDE integration. Contributed to the Arend plugin in IntelliJ IDEA.
- Developed an experimental LLM coding agent for writing Arend code using framework Koog.
- Fine-tuned models for Arend code completion.

Nov 2025 – Apr 2026

JetBrains, Research Intern in HoTT and Dependent Types Lab

Pafos, Cyprus

- I was developing the official library in the Arend language. I formalized different results in algebra and homotopy type theory, including following.
- Formalized Eckmann-Hilton argument, Eilenberg-MacLane spaces, Homologies of types.
- Formalized automorphisms of groups, Schur's Lemma, Maschke's Lemma, Group actions characterizations. (Some of these results are in the [1.10 release](#) [↗](#)).

June 2024 – Sept 2024

IPONWEB (acquired by Criteo), Machine Learning Intern

Pafos, Cyprus

- Criteo has a ML tool that analyzes sites and produces word2vec representations. Reverse-engineered the behaviour of this tool and analyzed its possible applications.

May 2023 – Sept 2023

Awards and Scholarships

Scholarship by "Rodnye Goroda" (a social investment program of PJSC "Gazprom нефт"), 2023-2024

JetBrains Scholarship in Neapolis University

Travel grant of Chebyshev laboratory for [Winter student school in mathematics and CS of HSE and MCS](#), Moscow

[Excellence Master Fellowship at Geneva University](#)

Teaching Experience

Sirius educational center , Teaching assistant on the course " Groups of intermediate growth "	Sochi, Russia Apr 2024 – May 2024
Laboratory for continuous mathematical education , mathematics teacher for gifted students	Saint-Petersburg, Russia 2021 – 2023
Mathematics and Computer Science faculty program for prospective students , Teaching assistant on the course Braid and knot theory	Saint-Petersburg, Russia June 2023

Selected Talks

On the geometry of free nilpotent groups Siberian summer conference	Novosibirsk, Russia 2021
"On the Poisson boundary of lamplighter groups" St. Petersburg Seminar on Representation Theory and Dynamical Systems	Saint-Petersburg, Russia
"Twisting numbers on braid and Thompson's groups" Topological Methods in Dynamics and Related Topics VII	N.Novgorod, Russia 2024
"Plus construction and simplicial methods" Conference "Algebraic Topology, geometry, combinatorics, and data analysis"	Pushkin, Russia Aug 2025
"The operad associated to a crossed simplicial group" Young Topologists Meeting 2026	Copenhagen, Denmark July 2026

Schools and Workshops

Teaching Higher Category Theory with Computers Topical workshop on directed HoTT and computer-formalized higher category theory, organized by Emily Riehl, Nima Rasekh, Jonathan Weinberger. Participant (travel and lodging funded).	ICERM, Brown University, Providence, RI Aug 2026
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Miscellaneous

- Finalist of 2020 Intel ISEF
- Intel ISEF alumni
- Winner of 2019 Baltic SEF, PDMI special prize in 2019 Baltic SEF
- 3rd team place in 2019 [International Tournament of Young Mathematicians](#) in Barcelona, Spain
- Organizer of the [Euler International Mathematical Institute's functional analysis seminar](#)
- English level C1 (IELTS 8.0/9, taken in 2020, 2024)
- Invited judge in Saint-Petersburg Tournament of Young Mathematicians (since 2021)
- Created mathematical problem for 2024 International Tournament of Young Mathematicians [10th in here](#)
- Invited judge in International Tournament of Young Mathematicians 2024, 2025
- [A member of Scientific Organization Committee in ETEAM 2025](#)